

INFORMATION & COMMUNICATION TECHNOLOGY
TECHNICIAN — NVQ LEVEL 4

MODULE M01

Maintaining Files & Folders

Use the Computer & Manage Files within Standard Operating Systems

Unit Code: K72S004U01 | NVQ Level: 02, 03 & 04

Duration: 24 Hours (Theory: 18 hrs | Practical: 6 hrs)

Developed in accordance with NAITA / TVEC National Competency Standards

NCS Learning Outcomes (ඉගෙනීමේ ඉලක්ක)

NCS document එකෙන් define කරන ලද competencies — ඔබ M01 නිමා කළ පසු කළ යුතු දේ:

After completing this module, you must be able to:

- ✓ Identify components of the Windows Desktop Environment
- ✓ Modify operating system configuration of a computer
- ✓ Handle files and folders according to given requirements
- ✓ Perform system maintenance to optimize computer performance
- ✓ Manage antivirus software and protect data

⚠ IMPORTANT: These 5 outcomes are the MINIMUM required to pass the M01 assessment.

1. Computer Start Up & Shut Down

1.1 Definition (අර්ථ දැක්වීම)

Start Up: Computer power on කරලා Operating System (OS) load කිරීමේ process එක.

Shut Down: Computer safely off කිරීමේ process එක.

Restart: Computer නවත්වා නැවත boot කිරීමේ process එක.

1.2 Standard Start Up Steps (ආරම්භ කිරීමේ ක්‍රමය)

1. UPS / Protected Power Supply switch on කරන්න.
2. Monitor power button press කරන්න.
3. CPU (System Unit) power button press කරන්න.
4. POST (Power On Self Test) screen display වෙනතෙක් රැඳෙන්න.
5. OS (Windows) load වෙනතෙක් wait කරන්න.
6. Login screen → Username & Password type කරන්න.
7. Desktop load වෙනතෙක් wait කරන්න.

🔑 Memory Trick: UPS → Monitor → CPU → Wait → Login → Desktop (U-M-C-W-L-D)

1.3 Standard Shut Down Steps (නිවා දැමීමේ ක්‍රමය)

8. Open applications සියල්ල save කරලා close කරන්න.
9. Start Menu click කරන්න.
10. Power icon click කරන්න.
11. 'Shut Down' select කරන්න.
12. Screen off වෙනතෙක් wait කරන්න.
13. Monitor power off කරන්න.
14. UPS power off කරන්න.

⚠ IMPORTANT: Applications close නොකර Shut Down කිරීම — data loss සිදුවිය හැකිය!

1.4 Power Options (Power Settings)

Power Option	Description
Sleep	RAM data save කරලා low power state — fast wake up
Hibernate	Disk එකට data save කරලා power off — slow wake up
Shut Down	Fully power off — data saved to disk
Restart	Shut down + Boot again — for updates/errors

1.5 Non-Responding Application (Hang වූ Program)

Application respond නොකළ විට Task Manager use කරන්න:

15. Ctrl + Alt + Delete press කරන්න.
16. 'Task Manager' click කරන්න.
17. Not responding program select කරන්න.
18. 'End Task' click කරන්න.

💡 Shortcut: Ctrl + Shift + Esc → Task Manager direct open

Exam Questions — Section 1

1. Computer startup process ලිවීමේ steps ලියන්න. (5 marks)
2. Shut down කිරීමේදී applications close නොකළොත් සිදුවන ගැටළු ලියන්න. (3 marks)
3. Sleep, Hibernate, Shut Down වල difference explain කරන්න. (6 marks)
4. Task Manager use කරන්නේ කෙසේද? Keyboard shortcut ද ලියන්න. (4 marks)

2. Basic Computer Configuration (Basic Settings)

2.1 System Information (System එකේ Details)

Control Panel → System → view system information:

- Processor type (CPU)
- RAM size
- Operating System name & version (e.g., Windows 11 Pro)
- Computer name
- System type (32-bit / 64-bit)

💡 Quick access: Windows Key + Pause/Break → System Properties window opens

2.2 Desktop Configuration

Setting	How to Access
Desktop Wallpaper	Right click Desktop → Personalize → Background
Date & Time	Right click Taskbar clock → Adjust date/time
Display Settings	Right click Desktop → Display Settings
Screen Resolution	Display Settings → Resolution
Color Settings	Display Settings → Color Profile
Taskbar Settings	Right click Taskbar → Taskbar Settings

2.3 Keyboard Language (Keyboard Language Change කිරීම)

19. Settings → Time & Language → Language & Region.
20. 'Add a Language' click කරන්න.
21. Preferred language (e.g., Sinhala) add කරන්න.
22. Keyboard layout select කරන්න.

 **Quick switch: Windows Key + Space → language switch**

Example: Sinhala typing සඳහා → 'Sinhala' language keyboard add කිරීම.

2.4 Install & Uninstall Software

Task	Steps
Install Software	Run installer (.exe/.msi) → Next → Accept → Install → Finish
Uninstall Software	Settings → Apps → Installed Apps → Select → Uninstall

 **IMPORTANT: Software install කිරීමේදී: License Agreement කියවන්න. Pirated software use ILLEGAL.**

2.5 Install Fonts

23. Download font file (.ttf or .otf).
24. Font file right-click කරන්න.
25. 'Install' හෝ 'Install for all users' click කරන්න.
26. Font MS Word, etc. දී appear වෙනවා.

2.6 Regional Settings

Settings → Time & Language → Region:

- Currency format (Rs. / \$)
- Date format (DD/MM/YYYY)

- Time format (12hr / 24hr)
- Number format (decimal/thousand separator)

💡 Example: Sri Lanka → Region → 'Sri Lanka' → Currency shows 'Rs.'

Exam Questions — Section 2

- ✍ 5. System Properties window එකේ display වෙත information 4ක් ලියන්න. (4 marks)
- ✍ 6. Keyboard language change කිරීමේ steps ලියන්න. (4 marks)
- ✍ 7. Software uninstall කිරීමේ steps ලියන්න. (4 marks)
- ✍ 8. Regional settings change කළ යුත්තේ ඇයි? Examples දෙකක් ලියන්න. (4 marks)

3. Windows Desktop Environment

3.1 Desktop Components (Desktop Elements)

Component	Description
Desktop	Main work area — icons & wallpaper
Taskbar	Bottom bar — open apps, Start button, clock
Start Menu	Windows icon click → all programs access
Icons	Shortcut images — files, folders, programs
Recycle Bin	Deleted files temporarily stored here
Notification Area	Right side of taskbar — clock, WiFi, battery
Search Bar	Type to search files and apps

3.2 Work with Icons (Icons හැසිරවීම)

- Select icon: Single click
- Open file/program: Double click
- Move icon: Click and drag
- Create shortcut: Right click file → Send to → Desktop (create shortcut)
- Arrange icons: Right click desktop → Sort by → Name/Size/Type/Date

3.3 Work with Windows (Windows කළමනාකරණය)

Action	Method
Minimize	Click _ button (window hides to taskbar)
Maximize	Click □ button (fills full screen)
Restore Down	Click □ button when maximized (smaller window)

Close	Click X button
Move Window	Click & drag title bar
Resize Window	Drag window edge/corner
Switch Windows	Alt + Tab / Click taskbar
Snap Window	Windows Key + Left/Right arrow

3.4 Window Arrangement (Windows සකස් කිරීම)

- Tile Vertically: Windows side by side
- Tile Horizontally: Windows top and bottom
- Cascade: Windows overlapping diagonally

Access: Right click Taskbar → select Tile/Cascade option

Exam Questions — Section 3

9. Windows desktop components 5ක් list කරන්න. (5 marks)
10. Window Minimize, Maximize, Close difference explain කරන්න. (3 marks)
11. Desktop icon shortcut create කරන method ලියන්න. (3 marks)

4. Files & Folders Management

4.1 File System Concept (File System Theory)

Computer හි data hierarchical structure එකක save කෙරේ:

Hierarchy (ඉහළ to පහළ):

Drive (C:\ D:\ E:\ USB Drive)

└─ Folder (e.g., Documents)

 └─ Sub-folder (e.g., School)

 └─ File (e.g., Assignment.docx)

4.2 Common File Types (File Extensions)

File Type	Extension
Word Processing	.docx .doc
Spreadsheet	.xlsx .xls .csv
Database	.accdb .mdb
Presentation	.pptx .ppt
Image	.jpg .png .bmp .gif

Audio	.mp3 .wav .aac
Video	.mp4 .avi .mkv
PDF Document	.pdf
Text File	.txt
Compressed File	.zip .rar
Executable	.exe .msi

⚠ IMPORTANT: File extension change කළොත් file open නොවිය හැකිය!

4.3 Create Folders & Sub-folders

27. Required location open කරන්න (e.g., D: drive).
28. Empty space right-click කරන්න.
29. 'New' → 'Folder' click කරන්න.
30. Folder name type කරන්න → Enter press.

💡 Shortcut: Ctrl + Shift + N → New folder (Windows 11)

4.4 File Operations (Copy, Move, Rename, Delete)

Operation	Method / Shortcut
Select single file	Single click
Select multiple (adjacent)	Click first → Shift + Click last
Select multiple (non-adjacent)	Ctrl + Click each file
Select all	Ctrl + A
Copy	Ctrl + C (or Right click → Copy)
Cut (Move)	Ctrl + X (or Right click → Cut)
Paste	Ctrl + V (or Right click → Paste)
Rename	F2 key (or Right click → Rename)
Delete	Delete key (goes to Recycle Bin)
Permanent delete	Shift + Delete (bypasses Recycle Bin)
Undo	Ctrl + Z
Search file	Windows Key + F or Search box

4.5 COPY vs MOVE — Key Difference

COPY (Ctrl+C + Ctrl+V)	MOVE (Ctrl+X + Ctrl+V)
Duplicate create වෙනවා	Original file move වෙනවා
Original file remain වෙනවා	Original file delete වෙනවා
Two copies exist	Only one copy remains
Use when: backup needed	Use when: reorganizing files

⚠ IMPORTANT: Copy = Duplicate. Move = Relocate. Exam question frequent!

4.6 Recycle Bin (Deleted Files)

- Delete key press → file goes to Recycle Bin
- Restore: Recycle Bin open → Right click file → Restore
- Permanently delete: Recycle Bin open → Right click → Delete
- Empty Recycle Bin: Right click Recycle Bin icon → Empty Recycle Bin

⚠ IMPORTANT: Shift + Delete = Bypass Recycle Bin = Cannot be recovered easily!

4.7 Search Files (Files Search කිරීම)

Method 1: File Explorer Search Box — folder open කරලා search box use.

Method 2: Windows Search — Windows Key + F press.

Search Filters:

- By Name: filename type
- By Date Modified: date range specify
- By Type: .docx, .jpg, etc.
- By Content: file inside text search
- Wildcards: '*' (any characters), '?' (single character)

💡 Example: *.jpg → all JPG files find කරයි. Report? → 'Report1', 'Report2' find.

Exam Questions — Section 4 🔄

- 📝 12. Copy සහ Move operations ලිය. Difference table format එකේ ලියන්න. (6 marks)
- 📝 13. File Extensions examples 5ක් ලියන්න. (5 marks)
- 📝 14. Recycle Bin uses ලියන්න. File restore කරන steps ලියන්න. (5 marks)
- 📝 15. Search file කිරීමේදී wildcards use කරන ආකාරය example සමඟ ලියන්න. (4 marks)

5. File & Folder Attributes

5.1 What are Attributes? (Attributes කිව්වොත්)

Attributes = File/Folder special properties — behaviour control කරයි.

5.2 Types of Attributes (Attribute වර්ග)

Attribute	Meaning
Read-Only	File view කළ හැකිය. Edit/Delete කළ නොහැකිය.

Hidden	File/Folder normal view කරන්නේ නෑ. 'Show hidden' enable කළ පමණකි show.
Archive	Backup software use. File changed since last backup = Archive ON.
System	OS system files. Delete/modify danger. Usually hidden.

5.3 How to Change Attributes (Attributes Change කිරීම)

31. File/Folder right-click.
32. 'Properties' click.
33. 'General' tab under 'Attributes' section.
34. Read-only / Hidden checkbox tick/untick.
35. Apply → OK.

 **Hidden files show කරන්න:** File Explorer → View → Show → Hidden Items tick.

Exam Questions — Section 5

 **16. File attributes 4 o's' නම් ලියා හා කෙටි විස්තරය ලියන්න. (4 marks)**

 **17. Read-Only attribute use කරන situation example ලියන්න. (2 marks)**

6. Compress & Extract Files (Files Compress කිරීම)

6.1 What is File Compression?

Compression = file size reduce කිරීම. Multiple files single archive file එකක save.

6.2 Common Formats

- .zip — Windows built-in support
- .rar — WinRAR software required
- .7z — 7-Zip software

6.3 Advantages (Compression Benefits)

- Storage space save
- Files email කිරීමේ size limit avoid
- Multiple files single archive
- Transfer speed increase

6.4 Steps — Compress (ZIP ගොනු)

36. Files/Folder select කරන්න.
37. Right-click.
38. 'Send to' → 'Compressed (zipped) folder' click.

39. New .zip file create වෙනවා.

6.5 Steps — Extract (Extract කිරීම)

40. ZIP file right-click.
41. 'Extract All' click.
42. Destination folder select.
43. 'Extract' click.
44. Files extract location show.

 **Double click ZIP → view contents. 'Extract' click → files out.**

Exam Questions — Section 6

 **18. File compression කිරීමේ advantages 4ක් ලියන්න. (4 marks)**

 **19. ZIP file compress කිරීමේ steps ලියන්න. (4 marks)**

7. System Maintenance (System Maintenance Tools)

7.1 Why Maintenance? (Maintenance ඇයි?)

Regular maintenance:

- Computer speed maintain කරයි
- Errors prevent කරයි
- Data protect කරයි
- System lifespan extend කරයි

7.2 Disk Error Checking (Disk Errors Check)

Hard disk errors scan කිරීම:

45. File Explorer open → Drive right-click.
46. 'Properties' → 'Tools' tab.
47. 'Error checking' → 'Check' click.
48. Errors found නම් fix කරනවා.

 **Command Prompt: chkdsk C: /f /r — thorough check**

7.3 Disk Cleanup (Disk Cleanup Tool)

Unnecessary files delete කිරීම:

- Temporary files
- Recycle Bin files
- Windows Update cleanup
- Browser cache

Access: Search → 'Disk Cleanup' → Drive select → OK → Clean up

7.4 Disk Defragmentation (Defrag)

Fragmented data rearrange කිරීම → disk access fast.

Access: Search → 'Defragment and Optimize Drives' → Optimize

⚠ IMPORTANT: SSD drives defrag නොකළ යුතුය! HDD drives only.

Fragmented Disk	Defragmented Disk
File parts scattered on disk	File parts stored together
Slow read/write speed	Fast read/write speed
More disk head movement	Less disk head movement

7.5 Backup & Restore

Backup = Data copy safe place save.

Backup Type	Description
Full Backup	All data copy. Slow but complete.
Incremental Backup	Changed files only. Fast.
Differential Backup	Changes since last full backup.

Windows Backup: Settings → Update & Security → Backup → Add a drive

⚠ IMPORTANT: Backup — data lost වෙනුවට කලින් කරන්න. Not after!

7.6 Manage Local Storage (Storage Management)

- Recycle Bin regularly empty
- Unused programs uninstall
- Temp files delete (C:\Windows\Temp)
- Storage Sense enable (Settings → System → Storage)

Exam Questions — Section 7

- 📝 20. System maintenance tasks 5ක් ලියන්න. (5 marks)
- 📝 21. Disk Defragmentation explain කරන්න. HDD/SSD difference ලියන්න. (4 marks)
- 📝 22. Backup types 3 explain කරන්න. (6 marks)

8. Antivirus Software Management

8.1 Computer Threats (Computer Threats)

Threat Type	Description
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Virus	Self-replicating program — files damage
Worm	Network spread — bandwidth consume
Trojan	Disguised as safe program — hidden attack
Spyware	User info secretly collect
Adware	Unwanted ads display
Ransomware	Files encrypt → money demand
Phishing	Fake websites/emails → login info steal

8.2 How Viruses Spread

- Infected USB / External drives
- Email attachments
- Downloaded files from internet
- Malicious websites
- Network shared folders

8.3 Antivirus Software

Antivirus software threats detect කරලා remove කරයි.

Popular Antivirus	Type
Windows Defender	Built-in (Windows)
Kaspersky	Third-party (Paid/Free)
Norton	Third-party (Paid)
Avast	Third-party (Free/Paid)
Bitdefender	Third-party (Paid)

8.4 Antivirus Settings & Scan

49. Antivirus software open.
50. 'Scan' option select.
51. Scan type select: Quick Scan / Full Scan / Custom Scan.
52. Scan start කරන්න.
53. Threats found → 'Remove' / 'Quarantine'.

Scan Types:

- Quick Scan — common threat locations only. Fast.
- Full Scan — entire computer. Slow but thorough.
- Custom Scan — specific drive/folder only.

8.5 Update Virus Definitions (Updates)

Virus definitions = database of known viruses.

Regular updates MUST — new viruses daily create.

- Automatic updates enable
- Manual update: Antivirus → Update → Check for updates

⚠ IMPORTANT: Outdated antivirus = New viruses detect නොකරයි = Danger!

8.6 Safe Computing Practices

- Unknown USB drives computer-ට connect නොකරන්න
- Suspicious email attachments open නොකරන්න
- Unknown websites download නොකරන්න
- OS regularly update
- Strong passwords use
- Public WiFi secure connections only

Exam Questions — Section 8

- 📝 23. Computer virus types 4ක් නම් ලියා describe කරන්න. (8 marks)
- 📝 24. Virus spread වන ආකාර 4ක් ලියන්න. (4 marks)
- 📝 25. Antivirus virus definition update කරන්නේ ඇයි? Explain කරන්න. (4 marks)
- 📝 26. Full Scan සහ Quick Scan ලිය difference. (4 marks)

9. Text Editing System (Notepad & Basic Editing)

9.1 Notepad — Basic Text Editor

Notepad = Windows built-in simple text editor.

- File → New: New file create
- File → Open: Existing file open
- File → Save As: Save with name/location/format
- File → Print: Print document

9.2 Basic Keyboard Shortcuts

Shortcut	Action
Ctrl + N	New file
Ctrl + O	Open file
Ctrl + S	Save
Ctrl + A	Select all text
Ctrl + C	Copy
Ctrl + X	Cut
Ctrl + V	Paste

Ctrl + Z	Undo
Ctrl + F	Find
Ctrl + H	Find & Replace
Ctrl + P	Print

💡 These shortcuts work in almost ALL Windows applications!

Exam Questions — Section 9 🧠

📝 27. Notepad use කරලා file create කරන steps ලියන්න. (3 marks)

📝 28. Keyboard shortcuts 5ක් ලියන්න. (5 marks)

10. Computer Laws (Legal Aspects)

10.1 Sri Lanka Computer Laws

Act / Law	What it Covers
Computer Crimes Act No. 24 of 2007	Hacking, unauthorized access, cyber crimes
Intellectual Property Act No. 36 of 2003	Software copyright, piracy
Payment Devices Frauds Act No. 30 of 2006	Online fraud, credit card fraud

⚠️ **IMPORTANT:** Using pirated software = violates Intellectual Property Act = Illegal!

Exam Questions — Section 10 🧠

📝 29. Sri Lanka computer-related laws 2ක් ලිය. (4 marks)

📝 30. Computer Crimes Act කෙසේ relevant ද? Example ලියන්න. (3 marks)

11. Worker Attitudes & Behaviors

NCS document ටි mention කළ workplace attitudes:

Attitude	Meaning
Responsibility	Assigned tasks properly complete
Self Esteem	Own ability value
Self Confidence	Work accurately without fear
Sociability	Co-workers with good relations maintain

Self Management	Own time & work manage
Integrity & Honesty	Truthful, no cheating
Punctuality	On time work/attendance
Cooperative	Others with work together
Self Motivated	Without being told — own initiative work

QUICK REVISION SUMMARY ⚡

A — Most Important Keywords

Term	Quick Definition
OS (Operating System)	Computer manage කරන software — Windows, macOS, Linux
File	Data store unit — .docx, .jpg, .pdf
Folder	Files organize කරන container
Drive	Storage device — C: D: USB
Recycle Bin	Deleted files temporary store
Antivirus	Virus detect & remove software
Virus Definition	Known virus database
Backup	Data copy safe place save
Defragmentation	HDD file parts rearrange — speed up
Compression	File size reduce
Attribute	File special property (Read-only, Hidden)
Task Manager	Running processes manage — Ctrl+Alt+Del

B — Key Shortcut Cheat Sheet 🖱️

Shortcut	Action
Ctrl + C	Copy
Ctrl + X	Cut/Move
Ctrl + V	Paste
Ctrl + Z	Undo
Ctrl + A	Select All
Ctrl + S	Save
Ctrl + F	Find

F2	Rename
Delete	Send to Recycle Bin
Shift + Delete	Permanent Delete
Ctrl + Alt + Del	Security screen / Task Manager
Ctrl + Shift + Esc	Task Manager direct
Windows Key + E	File Explorer open
Windows Key + D	Show Desktop
Windows Key + L	Lock Computer
Alt + Tab	Switch between windows
Ctrl + Shift + N	New Folder

C — Common Mistakes Students Make

Common Mistakes to Avoid:

- X Forgetting to SAVE work before shutdown
- X Confusing COPY vs MOVE — Remember: Copy = Duplicate, Move = Relocate
- X Shift+Delete bypasses Recycle Bin — cannot easily recover!
- X SSD drives should NOT be defragmented (only HDD)
- X Outdated antivirus = not protected against new viruses
- X File extension change = file may not open
- X Force shutdown without closing apps = data loss risk

D — NCS Competency Checklist

Can you do all these? (Exam ready check):

- ☐ Start and shut down computer safely
- ☐ Change date, time, display, language settings
- ☐ Create folders and sub-folders
- ☐ Copy, Move, Rename, Delete files
- ☐ Restore files from Recycle Bin
- ☐ Set Read-Only and Hidden attributes
- ☐ Compress and Extract ZIP files
- ☐ Run Disk Cleanup
- ☐ Check Disk Errors
- ☐ Run Antivirus scan
- ☐ Update Virus Definitions
- ☐ Create a backup

- ☐ Use Task Manager to end a non-responding app
- ☐ Search for files using wildcards

E — Full Exam Question Bank

SHORT QUESTIONS (2-3 marks each):

-  1. Define: File, Folder, Drive (3 marks)
-  2. List 4 file attributes. (4 marks)
-  3. Write 3 advantages of file compression. (3 marks)
-  4. Name 3 types of backup. (3 marks)
-  5. Name 4 types of malware. (4 marks)
-  6. Difference: Quick Scan vs Full Scan. (2 marks)
-  7. What is Recycle Bin? (2 marks)
-  8. Write 3 keyboard shortcuts. (3 marks)

STRUCTURED QUESTIONS (5+ marks each):

-  9. Computer startup procedure step-by-step write. (5 marks)
-  10. Copy and Move difference explain with examples. (6 marks)
-  11. Antivirus software importance explain. How to update? (5 marks)
-  12. System maintenance tasks 5 explain. (10 marks)
-  13. Disk Defragmentation explain with diagram/illustration. (5 marks)
-  14. File attributes 4 explain with examples. (8 marks)
-  15. Computer threats 4 identify and explain. (8 marks)